NUCLEAR POWER AND THE THREE-STAGE PROGRAMME

FISSION-TO-GRID CHAIN → CRITICALITY MILESTONE LADDER → CORE-CONTROL FUNCTION MAP → THERMAL-FAST COMPARISON → FISSILE-FERTILE CONVERSION MAP → THREE-STAGE PROGRAMME RAIL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW | ● SUBORDINATE EXTRA | ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 FISSION-TO-GRID CHAIN

FISSION-TO-GRID CHAIN | HEAVY NUCLEUS SPLITS AND RELEASES HEAT PLUS NEUTRONS | REMOVES CORE HEAT | STEAM SYSTEM | CONVERTS HEAT INTO WORKING-FLUID ENERGY | PRODUCES ELECTRICITY | REACTOR CAPACITY IS NOT ELECTRICITY GENERATED

STARTING CONDITION / INPUT	SCIENTIFIC OR ENGINEERING MECHANISM	OUTPUT / FAILURE BOUNDARY
FISSION → heavy nucleus splits and releases heat plus neutrons	STEAM SYSTEM → converts heat into working-fluid energy	RULE → reactor capacity is not electricity generated
COOLANT → removes core heat	TURBINE-GENERATOR → produces electricity	FISSION → heavy nucleus splits and releases heat plus neutrons

STARTING CONDITION / INPUT
- FISSION → heavy nucleus splits and releases heat plus neutrons
- COOLANT → removes core heat

SCIENTIFIC OR ENGINEERING MECHANISM
- STEAM SYSTEM → converts heat into working-fluid energy
- TURBINE-GENERATOR → produces electricity

OUTPUT / FAILURE BOUNDARY
- RULE → reactor capacity is not electricity generated
- FISSION → heavy nucleus splits and releases heat plus neutrons

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → reactor capacity is not electricity generated.

01

STAGE 01 CRITICALITY MILESTONE LADDER

CRITICALITY MILESTONE LADDER | CHAIN REACTION DECLINES | FIRST CRITICALITY | CONTROLLED SELF-SUSTAINING REACTION BEGINS | POWER ASCENSION | TESTED INCREASE TOWARD RATED CONDITIONS | GRID SYNCHRONISATION | GENERATOR CONNECTS TO GRID

SUBCRITICAL → chain reaction declines

FIRST CRITICALITY → controlled self-sustaining reaction begins → POWER ASCENSION → tested increase toward rated conditions → GRID SYNCHRONISATION → generator connects to grid

STARTING CONDITION / INPUT
- SUBCRITICAL → chain reaction declines
- FIRST CRITICALITY → controlled self-sustaining reaction begins

SCIENTIFIC OR ENGINEERING MECHANISM
- POWER ASCENSION → tested increase toward rated conditions
- GRID SYNCHRONISATION → generator connects to grid

OUTPUT / FAILURE BOUNDARY
- COMMERCIAL OPERATION → later declared operating status
- SUBCRITICAL → chain reaction declines

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POWER ASCENSION → tested increase toward rated conditions.

02

STAGE 02 CORE-CONTROL FUNCTION MAP

CORE-CONTROL FUNCTION MAP | CONTROL RODS | ABSORB NEUTRONS AND REGULATE REACTIVITY | SLOWS NEUTRONS IN A THERMAL REACTOR | TRANSFERS HEAT OUT OF THE CORE | BARRIER AGAINST RADIOACTIVE RELEASE | FOUR FUNCTIONS, FOUR DIFFERENT ANSWERS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
CONTROL RODS → absorb neutrons and regulate reactivity	COOLANT → transfers heat out of the core	RULE → four functions, four different answers
MODERATOR → slows neutrons in a thermal reactor	CONTAINMENT → barrier against radioactive release	CONTROL RODS → absorb neutrons and regulate reactivity

SYSTEM / DEVICE / INSTITUTION
- CONTROL RODS → absorb neutrons and regulate reactivity
- MODERATOR → slows neutrons in a thermal reactor

DECISIVE TECHNICAL DISTINCTION
- COOLANT → transfers heat out of the core
- CONTAINMENT → barrier against radioactive release

STATUS / UNIT / EVIDENCE LIMIT
- RULE → four functions, four different answers
- CONTROL RODS → absorb neutrons and regulate reactivity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOLANT → transfers heat out of the core.

03

STAGE 03 THERMAL-FAST COMPARISON

THERMAL-FAST COMPARISON | THERMAL REACTOR | MODERATED SLOWER-NEUTRON SPECTRUM | FAST REACTOR | NO MODERATOR AND FAST-NEUTRON SPECTRUM | HEAVY-WATER MODERATED STAGE-1 WORKHORSE | BREEDING OBJECTIVE WITH FERTILE BLANKET | COOLANT ALONE DOES NOT DEFINE THE SPECTRUM

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT

SYSTEM / DEVICE / INSTITUTION
- CONTROL RODS → absorb neutrons and regulate reactivity
- MODERATOR → slows neutrons in a thermal reactor

DECISIVE TECHNICAL DISTINCTION
- COOLANT → transfers heat out of the core
- CONTAINMENT → barrier against radioactive release

STATUS / UNIT / EVIDENCE LIMIT
- RULE → four functions, four different answers
- CONTROL RODS → absorb neutrons and regulate reactivity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COOLANT → transfers heat out of the core.

03

STAGE 03 THERMAL-FAST COMPARISON

THERMAL-FAST COMPARISON · THERMAL REACTOR · MODERATED SLOWER-NEUTRON SPECTRUM · FAST REACTOR · NO MODERATOR AND FAST-NEUTRON SPECTRUM · HEAVY-WATER MODERATED STAGE-1 WORKHORSE · BREEDING OBJECTIVE WITH FERTILE BLANKET · COOLANT ALONE DOES NOT DEFINE THE SPECTRUM

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
THERMAL REACTOR → moderated slower-neutron spectrum	PHWR → heavy-water moderated Stage-1 workhorse	RULE → coolant alone does not define the spectrum
FAST REACTOR → no moderator and fast-neutron spectrum	FBR → breeding objective with fertile blanket	THERMAL REACTOR → moderated slower-neutron spectrum

SYSTEM / DEVICE / INSTITUTION
- THERMAL REACTOR → moderated slower-neutron spectrum
- FAST REACTOR → no moderator and fast-neutron spectrum

DECISIVE TECHNICAL DISTINCTION
- PHWR → heavy-water moderated Stage-1 workhorse
- FBR → breeding objective with fertile blanket

STATUS / UNIT / EVIDENCE LIMIT
- RULE → coolant alone does not define the spectrum
- THERMAL REACTOR → moderated slower-neutron spectrum

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → coolant alone does not define the spectrum.

04

STAGE 04 FISSILE-FERTILE CONVERSION MAP

FISSILE-FERTILE CONVERSION MAP · FISSILE MATERIALS · NEUTRON CAPTURE · PU-239 PATHWAY · NEUTRON CAPTURE AND DECAY · U-233 PATHWAY · FERTILE MATERIAL · NEEDS IRRADIATION AND CONVERSION

U-235 / PU-239 / U-233 → fissile materials

U-238 → neutron capture → PU-239 pathway → TH-232 → neutron capture and decay → U-233 pathway → FERTILE MATERIAL → needs irradiation and conversion

STARTING CONDITION / INPUT
- U-235 / PU-239 / U-233 → fissile materials
- U-238 → neutron capture → PU-239 pathway

SCIENTIFIC OR ENGINEERING MECHANISM
- TH-232 → neutron capture and decay → U-233 pathway
- FERTILE MATERIAL → needs irradiation and conversion

OUTPUT / FAILURE BOUNDARY
- RULE → resource abundance is not ready reactor fuel
- U-235 / PU-239 / U-233 → fissile materials

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → resource abundance is not ready reactor fuel.

05

STAGE 05 THREE-STAGE PROGRAMME RAIL

THREE-STAGE PROGRAMME RAIL · STAGE 1 · PHWR, NATURAL URANIUM AND PLUTONIUM-BEARING SPENT FUEL · SEPARATES MATERIAL FOR THE BREEDER BRIDGE · STAGE 2 · FAST BREEDER AND FISSILE-INVENTORY MULTIPLICATION · THORIUM IRRADIATION · PREPARES U-233 ROUTE

STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel → REPROCESSING → separates material for the breeder bridge → STAGE 2 → fast breeder and fissile-inventory multiplication → THORIUM IRRADIATION → prepares U-233 route → STAGE 3 → long-term thorium-U-233 utilisation horizon

CONCEPT / ARCHITECTURE
- STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel
- REPROCESSING → separates material for the breeder bridge

MECHANISM / APPLICATION
- STAGE 2 → fast breeder and fissile-inventory multiplication
- THORIUM IRRADIATION → prepares U-233 route

READINESS / STATUS LIMIT
- STAGE 3 → long-term thorium-U-233 utilisation horizon
- STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STAGE 2 → fast breeder and fissile-inventory multiplication.

06

STAGE 06 PFBR STATUS CARD

PFBR STATUS CARD · IGCAR FAST-REACTOR AND SODIUM EXPERTISE · BHAVINI PROJECT RESPONSIBILITY · 500 MWE DESIGN VALUE IN FETCHED DAE TEXT · 6 APRIL 2026 · FIRST CRITICALITY ACHIEVED · NOT ESTABLISHED · GRID SYNCHRONISATION OR COMMERCIAL OPERATION

DESIGN → IGCAR fast-reactor and sodium expertise

BUILD / COMMISSION → BHAVINI project responsibility → RATING → 500 MWe design value in fetched DAE text → 6 APRIL 2026 → first criticality achieved

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

OUTPUT / FAILURE BOUNDARY

- REPROCESSING → separates material for the breeder bridge
- THORIUM IRRADIATION → prepares U-233 route
- STAGE 1 → PHWR, natural uranium and plutonium-bearing spent fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STAGE 2 → fast breeder and fissile-inventory multiplication.

06

STAGE 06 PFBR STATUS CARD

PFBR STATUS CARD | IGCAR FAST-REACTOR AND SODIUM EXPERTISE | BHAVINI PROJECT RESPONSIBILITY | 500 MWE DESIGN VALUE IN FETCHED DAE TEXT | 6 APRIL 2026 | FIRST CRITICALITY ACHIEVED | NOT ESTABLISHED | GRID SYNCHRONISATION OR COMMERCIAL OPERATION

DESIGN → IGCAR fast-reactor and sodium expertise

BUILD / COMMISSION → BHAVINI project responsibility → RATING → 500 MWe design value in fetched DAE text → 6 APRIL 2026 → first criticality achieved

STARTING CONDITION / INPUT
- DESIGN → IGCAR fast-reactor and sodium expertise
- BUILD / COMMISSION → BHAVINI project responsibility

SCIENTIFIC OR ENGINEERING MECHANISM
- RATING → 500 MWe design value in fetched DAE text
- 6 APRIL 2026 → first criticality achieved

OUTPUT / FAILURE BOUNDARY
- NOT ESTABLISHED → grid synchronisation or commercial operation
- DESIGN → IGCAR fast-reactor and sodium expertise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NOT ESTABLISHED → grid synchronisation or commercial operation.

07

STAGE 07 CLOSED FUEL-CYCLE LOOP

CLOSED FUEL-CYCLE LOOP | REACTOR IRRADIATION | CHANGES ISOTOPIC COMPOSITION | MANAGES HEAT AND RADIATION | SEPARATES REUSABLE FISSILE MATERIAL | REMOTE FABRICATION | PREPARES NEW FUEL | CLOSES MATERIAL ACCOUNTING LOOP

REACTOR IRRADIATION → changes isotopic composition → COOLING / HANDLING → manages heat and radiation → REPROCESSING → separates reusable fissile material → REMOTE FABRICATION → prepares new fuel → REUSE / WASTE → closes material accounting loop

STARTING CONDITION / INPUT
- REACTOR IRRADIATION → changes isotopic composition
- COOLING / HANDLING → manages heat and radiation

SCIENTIFIC OR ENGINEERING MECHANISM
- REPROCESSING → separates reusable fissile material
- REMOTE FABRICATION → prepares new fuel

OUTPUT / FAILURE BOUNDARY
- REUSE / WASTE → closes material accounting loop
- REACTOR IRRADIATION → changes isotopic composition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REUSE / WASTE → closes material accounting loop.

08

STAGE 08 INSTITUTIONAL SEPARATION

INSTITUTIONAL SEPARATION | GOVERNMENTAL POLICY AND STRATEGIC UMBRELLA | REACTOR AND FUEL-CYCLE R&D | COMMERCIAL CIVILIAN PLANT DEVELOPER AND OPERATOR | SAFETY REVIEW AND REGULATORY OVERSIGHT | PROMOTER, LABORATORY, OPERATOR AND REGULATOR DIFFER

DAE → governmental policy and strategic umbrella

BARC → reactor and fuel-cycle R&D → NPCIL → commercial civilian plant developer and operator → AERB → safety review and regulatory oversight

CONCEPT / ARCHITECTURE
- DAE → governmental policy and strategic umbrella
- BARC → reactor and fuel-cycle R&D

MECHANISM / APPLICATION
- NPCIL → commercial civilian plant developer and operator
- AERB → safety review and regulatory oversight

READINESS / STATUS LIMIT
- RULE → promoter, laboratory, operator and regulator differ
- DAE → governmental policy and strategic umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → promoter, laboratory, operator and regulator differ.

09

STAGE 09 FUEL-AND-PROJECT CHAIN

FUEL-AND-PROJECT CHAIN | URANIUM MINING AND MILLING | FUEL AND REACTOR-COMPONENT FABRICATION | MONAZITE, THORIUM AND RARE-EARTH PROCESSING LINK | FAST-REACTOR DESIGN AND SODIUM R&D | PFBR PROJECT EXECUTION AND OPERATION

UCIL → uranium mining and milling → NFC → fuel and reactor-component fabrication → IREL → monazite, thorium and rare-earth processing link → IGCAR → fast-reactor design and sodium R&D → BHAVINI → PFBR project execution and operation

STARTING CONDITION / INPUT
- UCIL → uranium mining and milling
- NFC → fuel and reactor-component fabrication

SCIENTIFIC OR ENGINEERING MECHANISM
- IREL → monazite, thorium and rare-earth processing link
- IGCAR → fast-reactor design and sodium R&D

OUTPUT / FAILURE BOUNDARY
- BHAVINI → PFBR project execution and operation
- UCIL → uranium mining and milling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IREL → monazite, thorium and rare-earth processing link.

09

STAGE 09 FUEL-AND-PROJECT CHAIN

FUEL-AND-PROJECT CHAIN | URANIUM MINING AND MILLING | FUEL AND REACTOR-COMPONENT FABRICATION | MONAZITE, THORIUM AND RARE-EARTH PROCESSING LINK | FAST-REACTOR DESIGN AND SODIUM R&D | PFBR PROJECT EXECUTION AND OPERATION

UCIL → uranium mining and milling → **NFC → fuel and reactor-component fabrication** → **IREL → monazite, thorium and rare-earth processing link** → **IGCAR → fast-reactor design and sodium R&D** → **BHAVINI → PFBR project execution and operation**

STARTING CONDITION / INPUT
- UCIL → uranium mining and milling
- NFC → fuel and reactor-component fabrication

SCIENTIFIC OR ENGINEERING MECHANISM
- IREL → monazite, thorium and rare-earth processing link
- IGCAR → fast-reactor design and sodium R&D

OUTPUT / FAILURE BOUNDARY
- BHAVINI → PFBR project execution and operation
- UCIL → uranium mining and milling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: IREL → monazite, thorium and rare-earth processing link.

10

STAGE 10 REACTOR TAXONOMY

REACTOR TAXONOMY | HEAVY WATER AND NATURAL-URANIUM STAGE-1 ROLE | LIGHT-WATER SYSTEMS WITH DIFFERENT STEAM CYCLES | FAST SPECTRUM AND BREEDING OBJECTIVE | SMALL-REACTOR LINES | DESIGN OR DEVELOPMENT STATUS | TYPE AND DEPLOYMENT STATUS MUST BE STATED TOGETHER

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PHWR → heavy water and natural-uranium Stage-1 role	FBR → fast spectrum and breeding objective	RULE → type and deployment status must be stated together
PWR / BWR → light-water systems with different steam cycles	AHWR / SMALL-REACTOR LINES → design or development status	PHWR → heavy water and natural-uranium Stage-1 role

SYSTEM / DEVICE / INSTITUTION
- PHWR → heavy water and natural-uranium Stage-1 role
- PWR / BWR → light-water systems with different steam cycles

DECISIVE TECHNICAL DISTINCTION
- FBR → fast spectrum and breeding objective
- AHWR / SMALL-REACTOR LINES → design or development status

STATUS / UNIT / EVIDENCE LIMIT
- RULE → type and deployment status must be stated together
- PHWR → heavy water and natural-uranium Stage-1 role

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → type and deployment status must be stated together.

11

STAGE 11 NUCLEAR ANSWER SPINE

NUCLEAR ANSWER SPINE | FISSION, CRITICALITY, FISSILE, FERTILE AND FUEL CYCLE | STAGE 1, PLUTONIUM BRIDGE, STAGE 2 AND THORIUM HORIZON | DAE BARC NPCIL AERB IGCAR BHAVINI AND FUEL CHAIN | CAPACITY CRITICALITY GRID GENERATION AND LEGAL STATUS | DATED FLEET FUEL REACTOR LAW COST AND SCHEDULE EVIDENCE

DEFINE → fission, criticality, fissile, fertile and fuel cycle → **TRACE → Stage 1, plutonium bridge, Stage 2 and thorium horizon** → **MAP → DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain** → **SEPARATE → capacity criticality grid generation and legal status** → **VERIFY → dated fleet fuel reactor law cost and schedule evidence**

CONCEPT / ARCHITECTURE
- DEFINE → fission, criticality, fissile, fertile and fuel cycle
- TRACE → Stage 1, plutonium bridge, Stage 2 and thorium horizon

MECHANISM / APPLICATION
- MAP → DAE BARC NPCIL AERB IGCAR BHAVINI and fuel chain
- SEPARATE → capacity criticality grid generation and legal status

READINESS / STATUS LIMIT
- VERIFY → dated fleet fuel reactor law cost and schedule evidence
- DEFINE → fission, criticality, fissile, fertile and fuel cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated fleet fuel reactor law cost and schedule evidence.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | POLICY, STRATEGIC DIRECTION AND OVERALL NUCLEAR-GOVERNANCE UMBRELLA. | R&D BACKBONE FOR REACTOR TECHNOLOGY, FUEL CYCLE AND ADVANCED | THE FAST-REACTOR AND SODIUM-TECHNOLOGY DESIGN CENTRE — IT DESIGNED | BHAVINI IS THE SEPARATE COMPANY THAT BUILT IT AND | CONFLATING IGCAR WITH BHAVINI IS A COMMON INSTITUTIONAL ERROR. | COMMERCIAL REACTOR OPERATOR/DEVELOPER FOR THE CIVILIAN FLEET | ALSO THE VEHICLE FOR THE BHARAT SMALL REACTOR (220

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- DAE: policy, strategic direction and overall nuclear-governance umbrella.
- BARC: R&D backbone for reactor technology, fuel cycle and advanced concepts (including the AHWR design and the BSMR line).
- IGCAR: the fast-reactor and sodium-technology design centre — it designed PFBR
- BHAVINI is the separate company that built it and will operate it.

READINESS, UNIT OR STATUS LIMIT
- Conflating IGCAR with BHAVINI is a common institutional error.
- NPCIL: commercial reactor operator/developer for the civilian fleet
- also the vehicle for the Bharat Small Reactor (220 MWe PHWR for captive industrial use) solicitation.
- BHAVINI: Stage 2 breeder-reactor implementation body linked to PFBR.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- NFC / UCIL / IREL: fuel fabrication, uranium mining, and monazite processing for
- thorium and rare earths — the physical supply chain beneath the three-stage narrative.
- AERB: technical-safety regulator, distinct from liability legislation — and, as of the verified position, not a statutorily independent authority .
- CAUTION: A major debate is how to balance ambitious capacity expansion with

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels